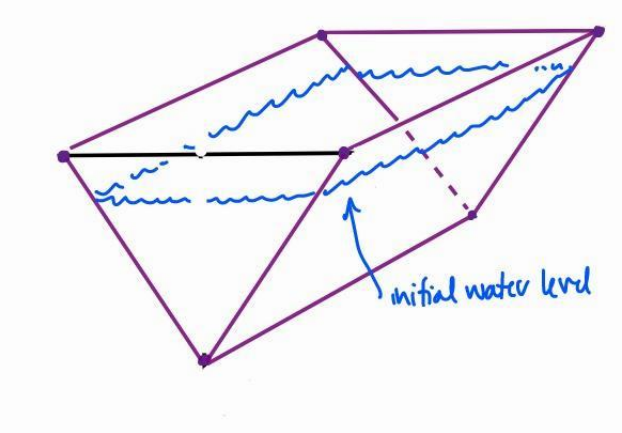
MATH 202 Your name: _____

Learning target A5, version 1

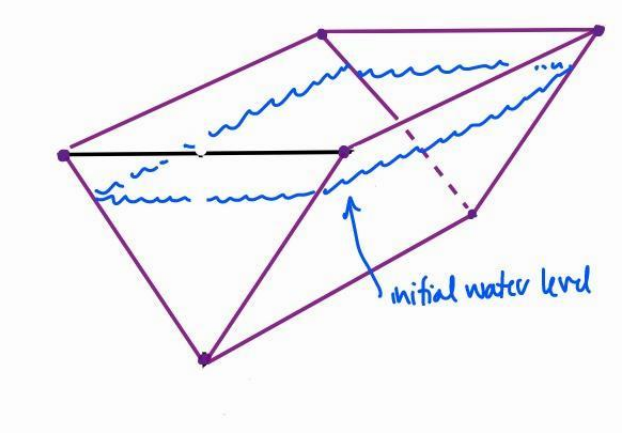
Use definite integrals to compute physical quantities, such as work, force, pressure, energy, etc.

Consider a trough with a triangular end as pictured below. The trough is 12 feet wide along its top, 15 feet deep, 18 feet long, and it is full of water to a depth of 13 feet. As usual, the weight density of water is 62.4 pounds per cubic foot.

1. *Determine the total amount of hydrostatic force on one end of the trough.* Clearly show relevant slices that go in to determining the definite integral that answers the question. State the appropriate definite integral clearly and evaluate it using technology to determine the total force. Include units on your result.



2. *Determine the total amount of work that would be required to empty the trough until 3 feet of water remains in it, while pumping the water to a truck whose tank is 6 feet above the top of the trough.* Clearly show the relevant slice(s) and computations that are involved in determining the definite integral that answers the question. State the appropriate definite integral clearly and evaluate it using technology to determine the total work. Include units on your result.



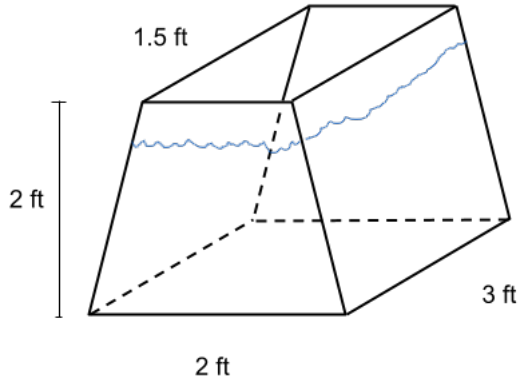
MATH 202 Your name: _____

Learning target A5, version 2

Use definite integrals to compute physical quantities, such as work, force, pressure, energy, etc.

Some asshole in a calculus problem constructed a trapezoidal fish tank as pictured below. The fish tank is 1.5 feet across the top, 2 feet across the bottom, 2 feet tall, and 3 feet long, and it's filled to a depth of 1.75 feet of water. As usual, the weight density of water is 62.4 pounds per cubic foot.

(a) *Determine the total amount of hydrostatic force on one of the flat ends of the tank.* Draw a slice, label it carefully, and show your work to write down the appropriate definite integral. Remember, in American units, pressure is 62.4 times water depth.



(b) *Find the total amount of work that would be required to pump all the water out of the tank.* Draw a slice, label it carefully, and show your work to write down the appropriate definite integral. Remember, in American units, "pounds" is a force.

